2026 NASA SPACE APPS CHALLENGE

INTERIM

ASTRONAUT HEALTH ASSESSMENT & DECISION-MAKING SIMULATOR

Making health decisions when communication with Earth is delayed.

THE IDEA

A web-based educational simulator where users experience astronaut health scenarios, assess simulated information and practise choosing a response.

TEAM A++

•LIM LI JING

LIM LI NING

JUSTIN YAP YING YI

NG XUAN YEE

LIVE DEMO | https://limlining.github.io/Nasa-Interim/

GITHUB | github.com/limlining/Nasa-Interim

Why astronaut health decisions are difficult

Spaceflight hazards can affect physical and psychological health, while distance from Earth can delay support.

01 SPACE RADIATION

02 ISOLATION & CONFINEMENT

03 DISTANCE FROM EARTH

04 GRAVITY FIELDS

05 HOSTILE / CLOSED ENVIRONMENTS

POTENTIAL HEALTH EFFECTS

Immune changes • cardiovascular risks • bone loss

Sleep disturbances • behavioural health concerns

THE CHALLENGE QUESTION

How can astronauts view health indicators, evaluate their condition and take appropriate action when immediate ground support is limited?

What is INTERIM?

A guided practice experience built around three essential steps in health management.

01 GATHER / VIEW

Displays simulated indicators such as heart rate, SpO_2 and blood pressure, alongside scenario information.

02 EVALUATE

Users review symptoms and indicator trends to assess the simulated astronaut's condition and urgency.

03 ACT

Users select a response and follow an action checklist within the scenario.

EXPERIENCE → ASSESS → DECIDE → REFLECT

Communication-delay and blackout modes let users practise decisions when ground guidance is not immediately available.

Explore six simulated astronaut health scenarios

Choose a situation, review the information and practise a response.

S1 CHEST PAIN

Chest pain + elevated heart rate

S2 INSOMNIA & LOW MOOD

Sleep problems + mood changes

S3 ELEVATED CO_2

High CO_2 + headache

S4 LOW BLOOD PRESSURE

Dizziness + low blood pressure

S5 POST-EVA SYMPTOMS

Nausea + skin redness after EVA

S6 IRREGULAR HEARTBEAT

Palpitations + irregular heartbeat

SIMULATION TOOLS

Changing indicators • trends • countdown • body budget

USER SUPPORT

Body navigation • daily check-in • Decision Log • Performance Profile

Demo scenario: Chest pain

The user reviews a simulated situation and chooses how to respond.

SCENARIO S1

Crushing chest pressure spreads to the left arm after exercise.

Simulated ground delay: 8 minutes

SIMULATED INDICATORS

HEART RATE

112 bpm

SpO_2

95%

BLOOD PRESSURE

148 / 92 mmHg

CHEST PAIN

6 / 10

FOUR ACTION CHOICES

01 Observe

02 Self-treat

03 Call crew

04 Emergency

The prototype compares the selected action with simulated reference guidance and records the decision.

Illustrative values and recommendations only — not clinical thresholds or medical advice.

From check-in to reflection

A repeatable decision loop supports practice before, during and after a simulated event.

01 CHECK IN

Sleep & readiness

02 SELECT

Scenario / body region

03 ASSESS

Symptoms & indicators

04 DECIDE

Choose one of 4 actions

05 WAIT / ACT

Delay or blackout

06 REVIEW

Compare, log & reflect

OBSERVE → CHOOSE → MONITOR → ESCALATE IF NEEDED → REFLECT

After a scenario, users receive simulated feedback and can revisit completed decisions in the Decision Log and Performance Profile.

Why INTERIM matters — and what comes next

An educational prototype for awareness and decision-making practice.

INTENDED IMPACT

- Builds awareness of spaceflight health risks
- Practises assessment of simulated indicators
- Encourages timely decisions and crew escalation
- Supports reflection

CURRENT LIMITATIONS

- Uses simulated health data and ground advice
- No real-time sensor integration
- No clinically validated decision rules
- Not for diagnosis or treatment

FUTURE DEVELOPMENT

- Explore relevant NASA health datasets
- Integrate approved wearables
- Add more indicators and trend analysis
- Validate scenarios with experts

SUPPORT CREW JUDGEMENT — NEVER REPLACE THE ASTRONAUT OR MEDICAL TEAM.

LIVE DEMO https://limlining.github.io/Nasa-Interim/

GITHUB https://github.com/limlining/Nasa-Interim/ • Thank you